VERIFICATION CORE REPORT

Topic: **Python 3.12 String Processing** | College: **DPU**
Tech: **Python 3.12** | Professor: **Prof. Report Available** | Deck ID: **test_rep** | Date: 2026-10-17 12:54:53
Docker Image Digest:
python:3.12-slim@sha256:4a8b21cd839e0123ef456789abcde0123456789abcdef0123456789abcdef01

Metric	Value	Metric	Value
Found vs Executed	19 found / 19 run	Unverified (Not Run)	0
Snippets Verified	0/19	Auto-Healed	0
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	10000%
Template QA Passed	10/10	Docker Backend Runs	5

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 2	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	64.7294521331787 ms	`Hello, world!`
Slide 3	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	50.8236885070800781 ms	`olleH`
Slide 4	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	38.1077233154296875 ms	`Result: Verified Output`
Slide 5	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	57.0811546325683594 ms	`(none)`
Slide 6	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	56.2857902526855469 ms	`(none)`
Slide 7	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	49.0476657348632812 ms	`Result: Verified Output`
Slide 8	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	57.5024155120849609 ms	`(none)`

Slide 9	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	60.1827583007812	Hello, world! hello, world! HELLO, WORLD! Hello, World! hELLO, wORLD!
Slide 10	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	65.2021199798584	Original: Hello, World!, Updated: HELLO, WORLD!
Slide 3	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE	-	(none)
Slide 4	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE	-	(none)
Slide 5	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE	-	(none)
Slide 6	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE	-	(none)
Slide 8	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE	-	(none)
Slide 3	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	203.80476760864258	Advanced String Manipulation in Python 3.12 Executed in Python 3.12
Slide 4	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	201.45692550659180	Python 3.12 String Processing: Concept 2 Executed in Python 3.12
Slide 5	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	197.52589233398438	Advanced String Methods in Python 3.12 Executed in Python 3.12
Slide 6	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	203.67015075683594	Advanced String Methods in Python 3.12 Executed in Python 3.12
Slide 8	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	199.73694595336914	Advanced String Methods in Python 3.12 Executed in Python 3.12

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-5-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
STRETCH	EX-7-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning`

main.py
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 3 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

EX-5-MEDIUM (MEDIUM) - Sandbox Execution Runs:
- **reference**: PASS
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 2 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

EX-7-STRETCH (STRETCH) - Sandbox Execution Runs:
- **reference**: PASS
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

3. Template Compliance QA

Slide	Compliance Status	Details

Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.
Slide 6	COMPLIANT	All bounding boxes and fonts compliant.
Slide 7	COMPLIANT	All bounding boxes and fonts compliant.
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.
Slide 9	COMPLIANT	All bounding boxes and fonts compliant.
Slide 10	COMPLIANT	All bounding boxes and fonts compliant.